

Scalability & Future Plan

Project Demonstration · AI/ML and GenAI Track

Scalability

- Replace SQLite with PostgreSQL by changing DATABASE_URL (SQLAlchemy already supports this).
- Run several Uvicorn/Gunicorn workers behind a reverse proxy (Nginx or Caddy) with HTTPS.
- Move Gemini calls to background jobs or async requests so slow AI calls do not block workers.
- Add caching and rate limiting to control API cost.
- Deploy the Docker image to a cloud service with a managed database.

Future plan

- Secure all admin routes and add user accounts and login.
- Progress tracking: mark workouts complete and log weight over time.
- Structured (JSON) plan output for richer display, calendars and reminders.
- Optional wearable / step-count integration.
- Multi-language support and a mobile-friendly app.
- Automated test suite (pytest) and continuous integration on GitHub.